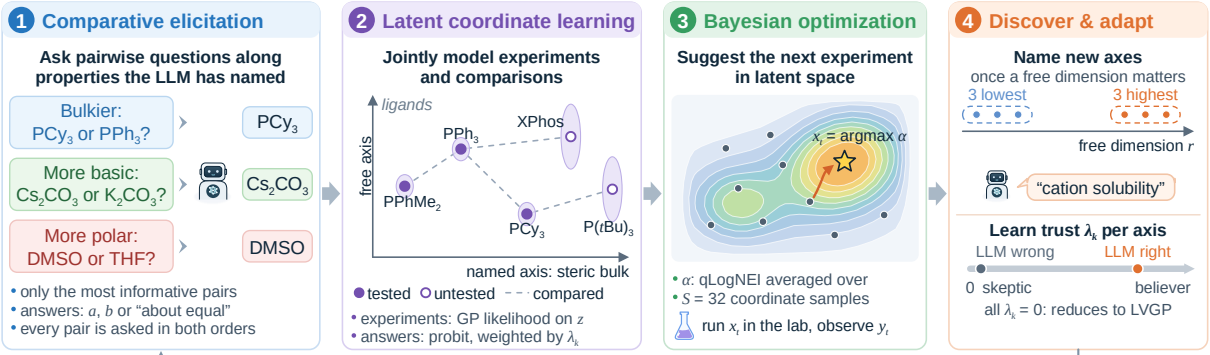

1 Comparative elicitation
Ask pairwise questions along properties the LLM has named
Bulkier: PCy3 or PPh3?
PCy3
More basic: Cs2CO3 or K2CO3?
Cs2CO3
More polar: DMSO or THF?
DMSO
• only the most informative pairs
• answers: a, b or "about equal"
• every pair is asked in both orders
2 Latent coordinate learning
Jointly model experiments and comparisons
ligands
free axis
PPh3
XPhos
PPhMe2
PCy3
P(tBu)3
named axis: steric bulk
tested
untested
compared
• experiments: GP likelihood on z
• answers: probit, weighted by λk
3 Bayesian optimization
Suggest the next experiment in latent space
xt = argmax α
• α: qLogNEI averaged over
• S = 32 coordinate samples
run xt in the lab, observe yt
4 Discover & adapt
Name new axes
once a free dimension matters
3 lowest
3 highest
free dimension r
"cation solubility"
Learn trust λk per axis
LLM wrong
LLM right
0 skeptic
believer
all λk = 0: reduces to LVGP
next iteration: add (xt, yt), refit, ask the most informative pairs